Joao Guilherme Lopes

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SUMMARY

An Embedded Software Engineer with 2+ years of experience developing real-time production-grade systems, proficient in multiple programming languages. Experienced in hardware-software co-design, low-level firmware and modular system integration, with a proven ability to deliver robust embedded applications, enhance system reliability and streamline processes in fast-paced environments. Privileged to have learned and collaborated with senior software and electronics colleagues that led me to become confident and fully autonomous in my work.

EXPERIENCE

Borg&Overström **June 2022 – Present, Norwich**

Embedded Software Engineer

- Led software development for embedded systems, supporting end-to-end product progress, OEM integrations, prototyping and system testing.
- Created bare-metal firmware and peripheral drivers ensuring precise timing and efficiency, low-level peripheral control and resource usage.
- Integrated communication interfaces into embedded systems, including BLE, MQTT, I2C, SPI, CAN, UART, MODBUS, RS232, RS485, among others.
- Manually assembled PCBs at component level for rapid prototyping.
- Extensive PCB and product diagnostics across all development stages.
- Reduced customer support tickets by 60% by improving firmware and implementing systematic bug fixes, leading to enhanced product reliability and decreased returned products.
- Implemented automated test systems for production line, reducing manual errors and increasing efficiency by 30%.
- Designed and optimized hardware schematics for both production-ready and prototype systems using EAGLE, Altium and Fusion 360.
- Developed automation for microcontrollers (MCU) flashing into production lines, streamlining programming and configuration, saving 2% to 4% suppliers costs and improving traceability.
- Engaged closely with other departments to ensure component availability, resolve support issues and maintain high customer satisfaction.
- Collaborated with colleagues and external companies on product certifications such as UL, FCC, NSF and Energy Star to meet relevant industry standards, enabling sales on different countries.

PROJECTS

Enhanced Flavour Machine (NRF/NXP/ESP32)

- Developed complete firmware set for flavoured water dispenser with precise real-time control of pumps, valves, sensors, Bluetooth, EEPROM, touch TFT display and scalable system extensions.
- Designed a modular architecture to support external components such as IoT, UV filtration, pay gate, external hot tanks, LED controls and multiple communication protocols.
- Contributed to IoT framework with MQTT, Modbus and UART and cellular communication framework to enable real-time data exchange.
- Researched system requirements and engaged with suppliers to select the most suitable components.
- Coded an animated Graphical User Interface (GUI) for a Human-Machine Interface (HMI) display software over two-way MIPI protocol.

Automated Test Station

- Developed an automated production-line test system to verify machine operation, sensors functionality, communication modules (I2C, UART, BLE) and control outputs.
- Designed co-existing hardware and software tailored to production requirements, enabling seamless integration with existing systems.
- Reduced manual testing by implementing automated diagnostic routines, resulting on speeding up production lines by 37.5%.
- Decreased escaped defects, improving overall reliability and reducing post-deployment failures.

Bluetooth Low Energy

- Developed cross-platform Android/iOS applications to control unit dispensing options and collect real-time operational data.
- Engineered a Bluetooth Low Energy (BLE) communication network between multiple boards, enabling intelligent coordination and status sharing.
- Designed logic for automatic rerouting to the nearest available unit based on signal strength (RSSI), improving user experience and system efficiency.
- Delivered seamless BLE integration compatible across the entire product range, supporting both current and future system variants.

Firmware Flashing System

- Developed a Linux-based firmware flashing tool using Python, SQLite and J-Link, enabling cross-platform MCU programming in production lines.
- Built a configurable API for real-time communication between flashing tools and operators, supporting dynamic firmware selection and flexibility
- Enhanced traceability via SQLite, logging important data such as barcodes, firmware version, flash counts, MCU targets and unit ID for improved quality control.

Room Conditioning Automation System (STM32)

- Designed and implemented a centralized room condition automation system for controlling and regulating environmental elements with both automatic and user-configurable modes.
- Developed a versatile GUI system with an intuitive user interface, enabling broad user control and a smooth interaction experience.
- Designed a PCB incorporating all necessary hardware components, ensuring seamless integration with the embedded system for optimal performance.

EDUCATION

Coventry University | Coventry

BEng in Computer Hardware and Software Engineering - Upper Second Class (2:1) **Sep 2019 – Jan 2023**

Foundation Engineering - Upper Second Class (2:1) **Sep 2018 – Jul 2019**

SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, HTML, VHDL, Assembly

Embedded Systems: Nordic, NXP, ITE, ESP-32, STM32, Raspberry Pi, Arduino

Communication Protocols: I2C, SPI, UART, SWD, BLE, MQTT, CAN, RS232, RS485, MODBUS, MIPI

Hardware Design: EAGLE, Fusion360, Altium Designer

Development Tools: Visual Studio Code, GitHub, Flutter, IntelliJ, Xcode, Android Studio, Pico Log, MATLAB, Proteus, J-Link

Platforms: Mac OS, Microsoft Windows, Linux (Ubuntu, Slackware, Debian), Android

CERTIFICATIONS

Photoshop Certification - Coventry University **March 2022**

Java Programming - Duke University (Coursera) **June 2021**

Cyber Security - Coventry University **December 2020**